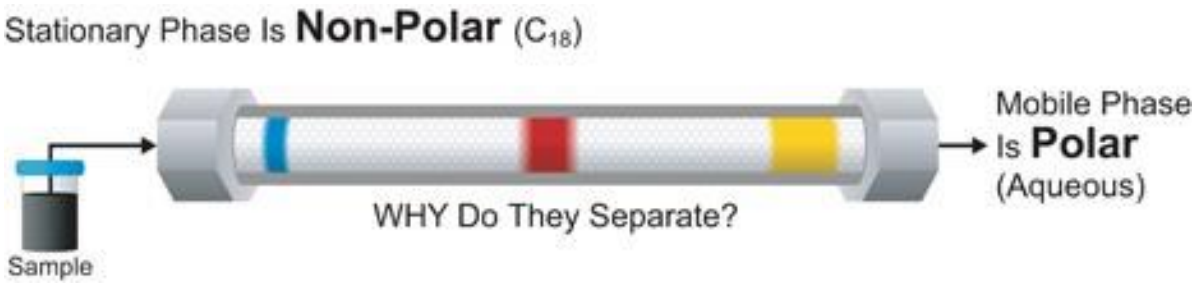
의약품분석학II
- Lecture 6 -

의약품기기분석학

제2장. 분리분석법

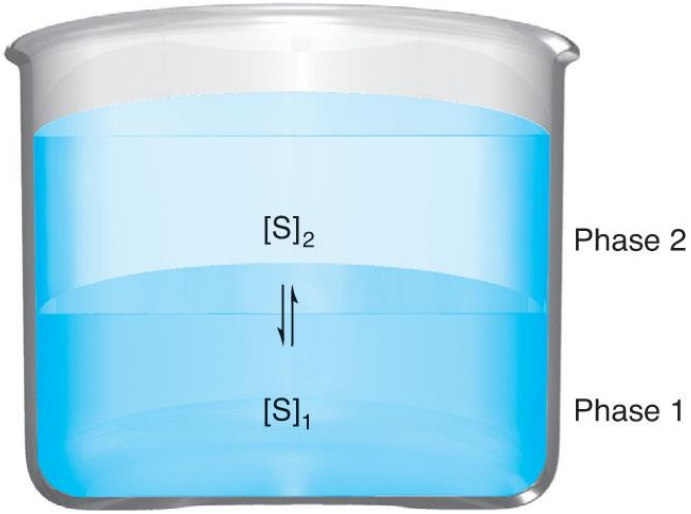
2.1 서론

2.2 크로마토그래피의 기초원리



Solvent extraction

- **Extraction is the transfer of a solute from one phase to another**
- **Extraction is carried out to isolate or concentrate the desired analyte or to separate it from interfering species**



Partition coefficient, *K*, is the equilibrium constant for a partition reaction

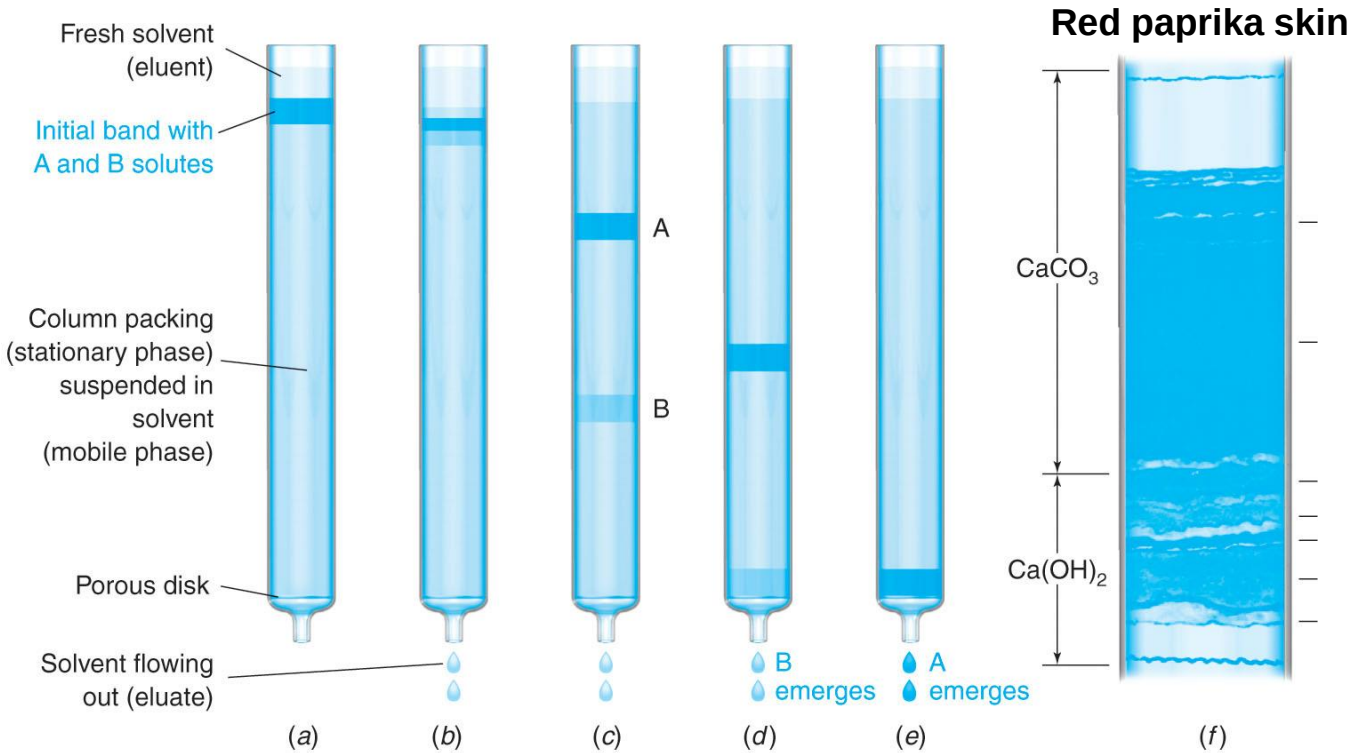
S (in phase 1) ⇆ S (in phase 2)

$$K = \frac{A_{S_2}}{A_{S_1}} \approx \frac{[S]_2}{[S]_1}$$

Liquid-liquid extraction (LLE)

What is chromatography?

- **Chromatography = "*chromatos*" (color) + "*graphein*" (to write) = color writing**
- **Chromatography works the same way as extraction, but one phase is held in place (i.e. stationary phase, 고정상) while the other moves past it (i.e. mobile phase, 이동상): transfer (or partitioning) of a solute between the stationary phase and the mobile phase**



Columns

- **Packed**
- **Open tubular**

Elution(용리): process of passing liquid or gas through chromatography column

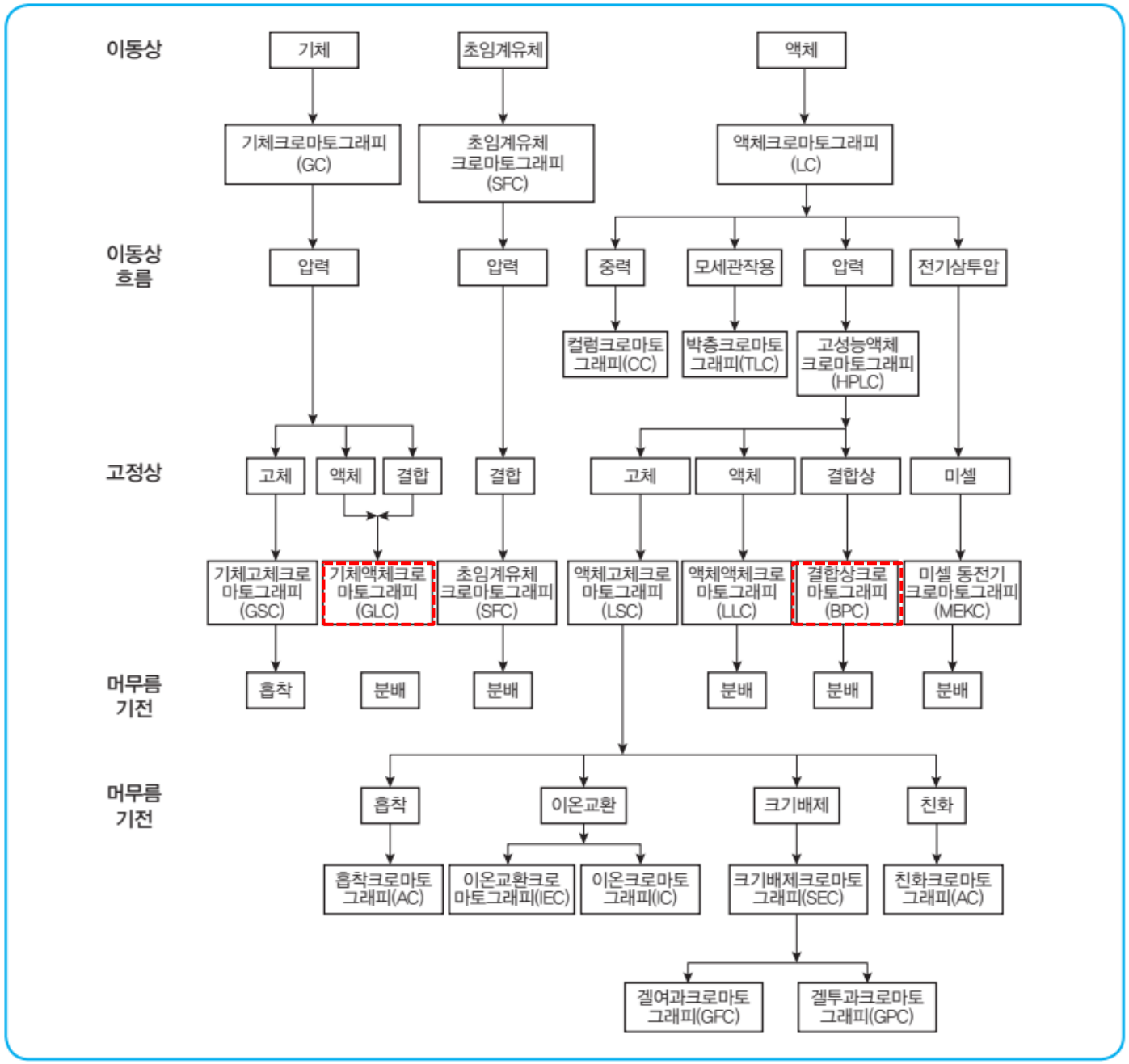
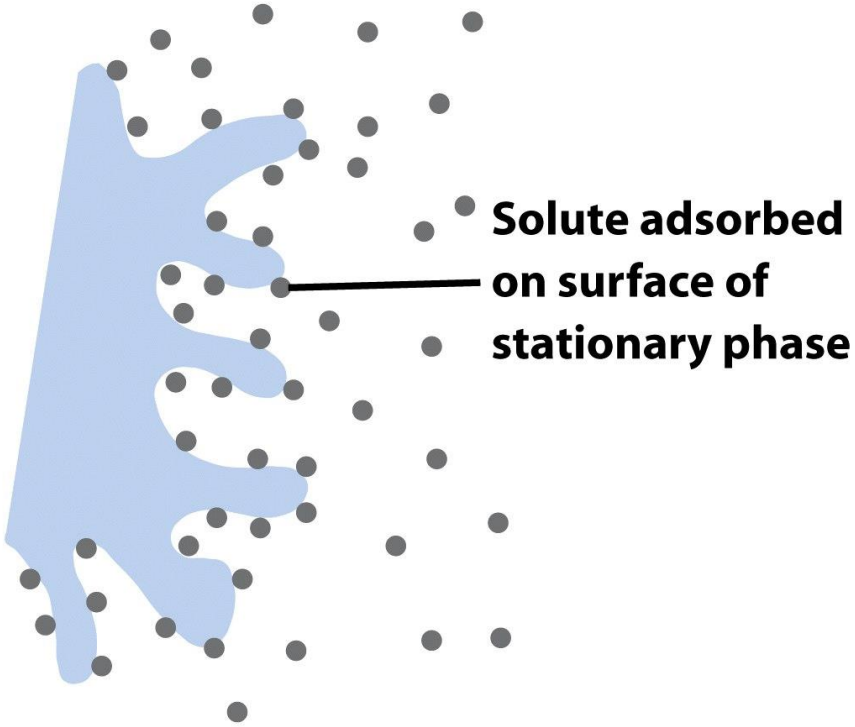


그림 2-6 크로마토그래피의 분류

Types of chromatography

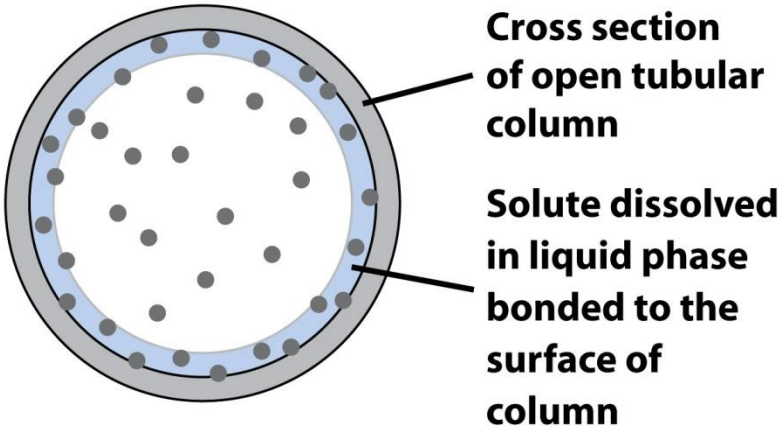
흡착크로마토그래피(Adsorption chromatography)



- The oldest type of chromatography
- Solid stationary phase and liquid/gas mobile phase
- Solute is adsorbed on the surface of the solid particles (stationary phase)
- Normal and reversed-phase HPLC (Silica gel, bonded silica, alumina etc.)

Types of chromatography

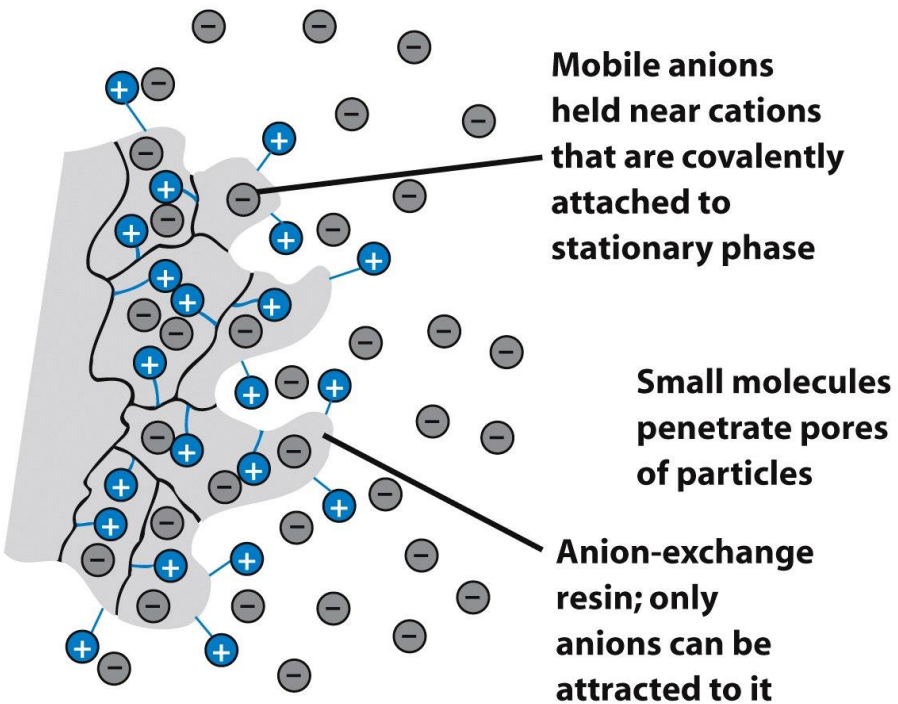
분배크로마토그래피(Partition chromatography)



- Liquid stationary phase and liquid/gas mobile phase
- Silica (SiO_2) column for gas chromatography: liquid stationary phase bonded to solid surface, which is inside of the silica column
- Solute equilibrates between the stationary liquid and the mobile phase (a flowing gas in gas chromatography)

Types of chromatography

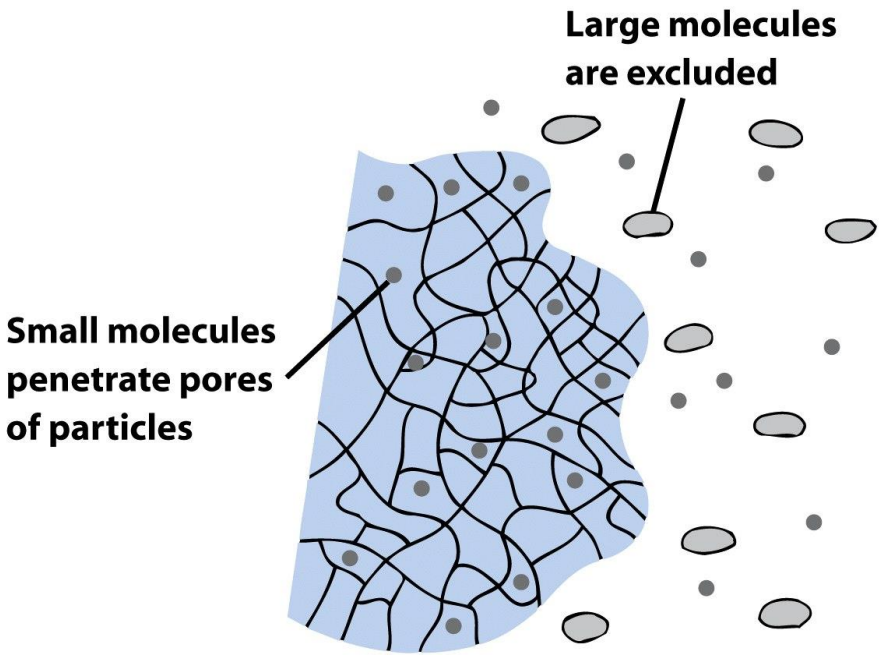
이온교환크로마토그래피(Ion-exchange chromatography)



- Solid stationary phase (ion-exchange resin) and liquid mobile phase
- Anions, e.g. $-SO_3^-$ (for cation-exchange resin) or cations, e.g. $-N(CH_3)_3^+$ (for anion-exchange resin) are covalently attached to the stationary solid phase (a resin)
- Solute ions of the opposite charge are attracted to the stationary phase
- Retention is based on the attraction between solute ions and charged sites bound to the stationary phase

Types of chromatography

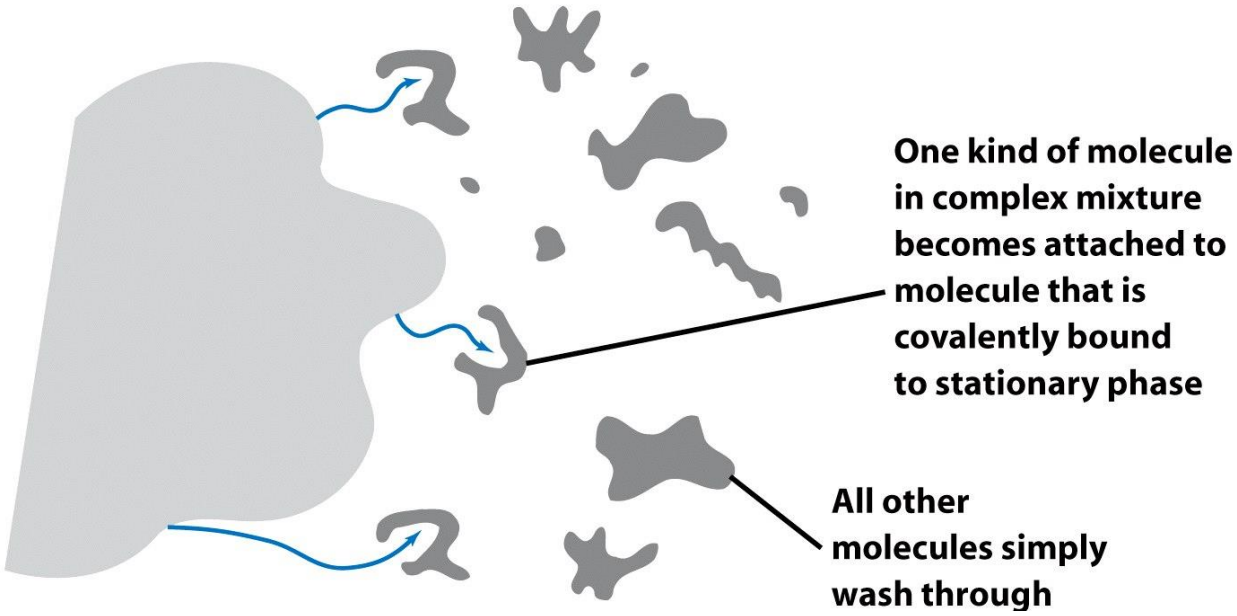
크기배제크로마토그래피[Molecular exclusion (size exclusion, gel filtration or gel permeation) chromatography]



- Solid stationary phase and liquid mobile phase
- Separates molecules by size: large molecules pass through the column faster than small molecules
- Small molecules penetrate the pores in the stationary phase, but large molecules do not; small molecules must pass through a larger volume than large molecules
- Widely used in biochemistry to purify macromolecules (e.g. proteins)
- Desalting: salts of low molecular mass (or any small molecule) can be removed from solutions of large molecules

Types of chromatography

친화크로마토그래피(Affinity chromatography)



- **Based on specific interaction between solute molecule and a molecule that is covalently attached (immobilized) to the stationary phase**
- **Most selective kind of chromatography used to isolate a single compound from a complex mixture, e.g. interactions between antigens (immobilized molecule) and antibodies (solute), enzymes and substrates, receptors and ligands**

A plumber's view of chromatography

Flow of liquid or gas through a pipe!

≈ Flow of mobile phase through a column!

Volume flow rate(부피유속), u_v (mL/min)

- volume of solvent per unit time traveling through column

Linear flow rate(길이유속), u_x (cm/min)

- distance per unit time traveled by solvent

$$u_x\ (\text{cm/min}) = \frac{u_v\ (\text{mL/min})}{\text{volume per unit column length (mL/cm)}}$$

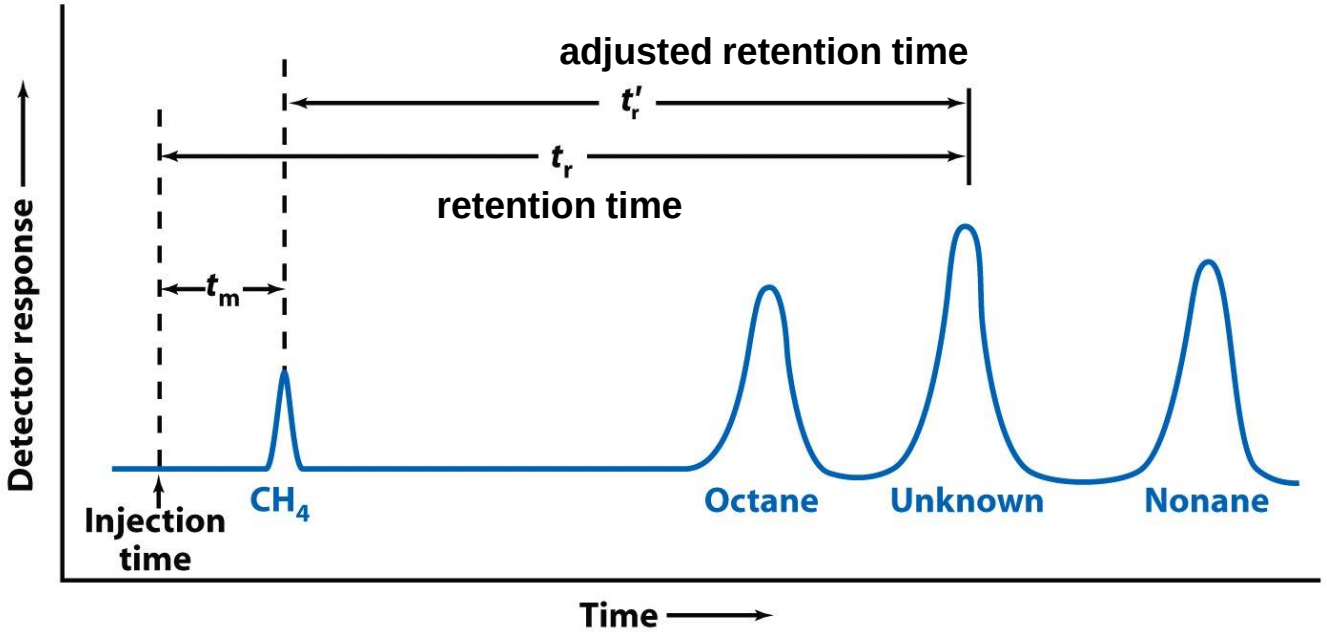
$$= \frac{u_v\ (\text{mL/min})}{(\pi r^2 \times 1)\ (\text{mL/cm})} \quad \text{(for open tubular columns)}$$

(r : inner radius of column)

A plumber's view of chromatography

The chromatogram

A graph showing the detector response as a function of elution time



t_r : retention time, 유지시간 또는 머무름시간 (time needed for an analyte to reach detector after injection)

t_m (t_0, dead time or void time): time needed for unretained mobile phase to reach detector after injection; 용질이 이동상에 머문 시간

t_r': adjusted retention time, $t_r' = t_r - t_m$; 용질이 고정상에 머문 시간

A plumber's view of chromatography

The chromatogram

Relative retention (separation factor) 분리인자, α :

$$\alpha = \frac{t'_{r2}}{t'_{r1}} \qquad t'_{r2} > t'_{r1} \qquad \alpha > 1$$

Relative retention is fairly independent of flow rate

Unadjusted relative retention, γ :

$$\gamma = \frac{t_{r2}}{t_{r1}}$$

Retention (or Capacity) factor 용량인자, k' :

$$k' = \frac{t_r - t_m}{t_m} = \frac{t'_r}{t_m}$$

- Adjusted retention time expressed in multiples of t_m (용질이 고정상에 머무는 시간(t_r')을 이동상에 머무는 시간(t_m)의 배수로 표현한 것)
- The longer a component is retained by a column, the greater is the retention factor.

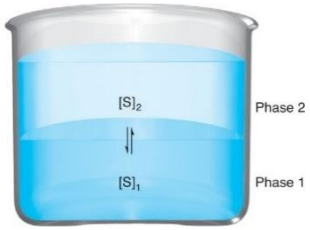
A plumber's view of chromatography

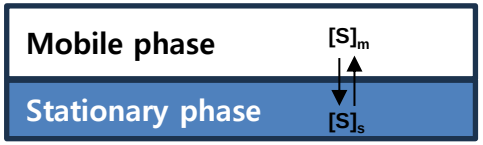
Relation between retention time and the partition coefficient

$$k' = \frac{t_r'}{t_m} = \frac{\text{time solute spends in stationary phase}}{\text{time solute spends in mobile phase}} = \frac{\text{moles of solute in stationary phase}}{\text{moles of solute in mobile phase}}$$

$$k' = \frac{c_s V_s}{c_m V_m} = K\frac{V_s}{V_m} = \frac{t_r - t_m}{t_m} = \frac{t_r'}{t_m}$$

$$k' \propto K \propto t_r'$$





Relative retention time, $\alpha = \frac{t_{r2}'}{t_{r1}'} = \frac{k_2'}{k_1'} = \frac{K_2}{K_1}$

Retention volume, V_r : the volume of mobile phase required to elute a particular solute from the column

$$V_r = t_r \cdot u_v$$

u_v : volume flow rate

V_r of a particular solute is constant over a range of flow rate

Q: 다음 각각의 경우 용질이 고정상에 머무는 시간은 이동상에 머무는 시간의 몇 배인가?

- $k'=0$
- $k'=1$
- $k'=5$

Efficiency of separation

Factors contributing to how well compounds are separated by chromatography:

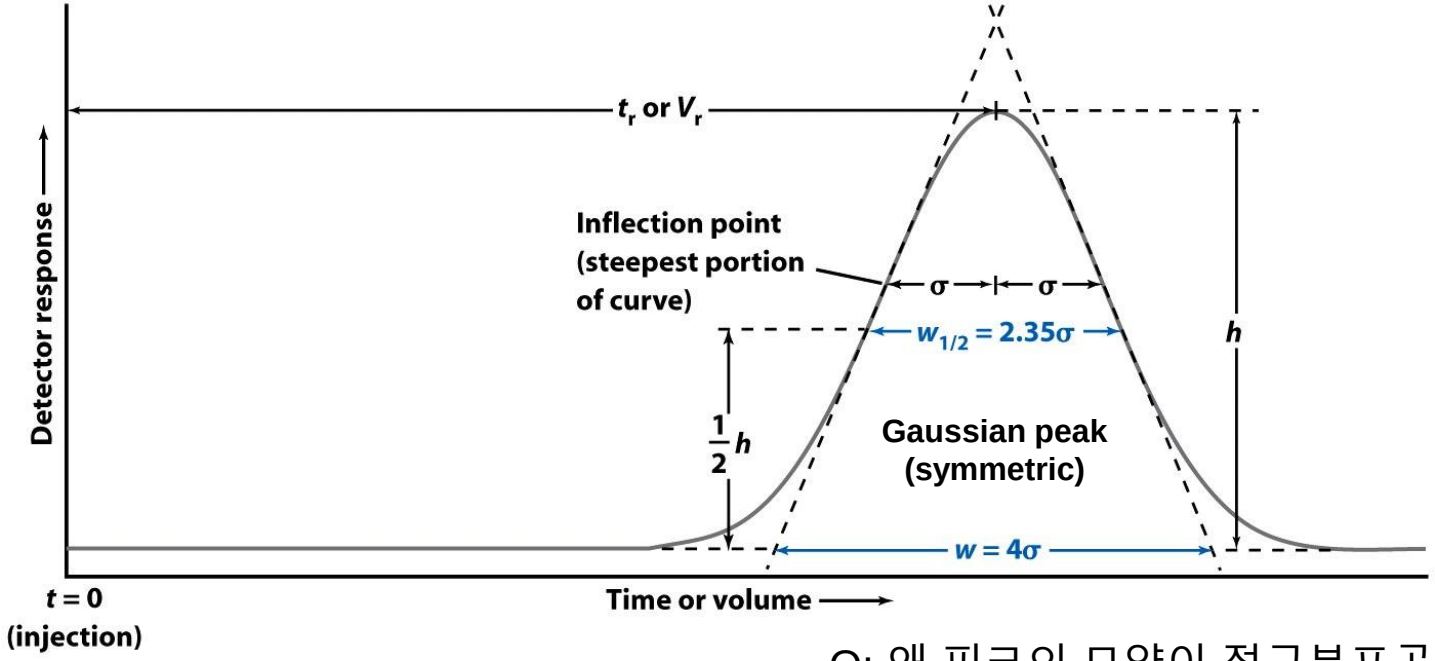
- **Difference in retention times between peaks**
- **Peak width**

$$\text{Resolution(분리도)}, R = \frac{\Delta t_r}{w_{av}} = \frac{\Delta V_r}{w_{av}} = \frac{\Delta t_r}{1.70 w_{1/2av}} = \frac{0.589 \Delta t_r}{w_{1/2av}}$$

$$w = 4\sigma$$

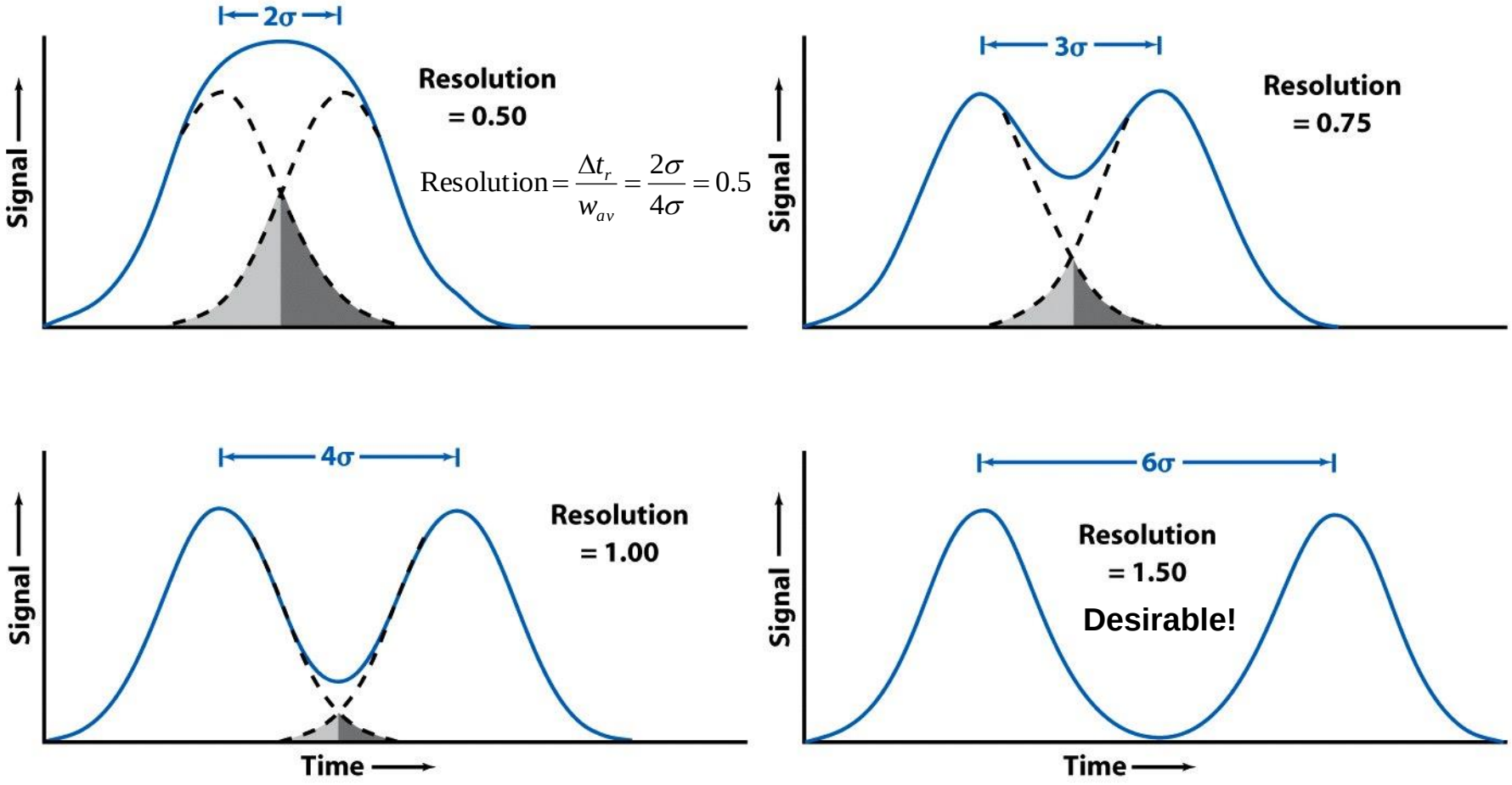
$$w_{1/2} = 2.35\sigma \qquad \sigma = w_{1/2}/2.35$$

$$w = 4 \times \frac{w_{1/2}}{2.35} = 1.70 w_{1/2}$$



Q: 왜 피크의 모양이 정규분포곡선일까?

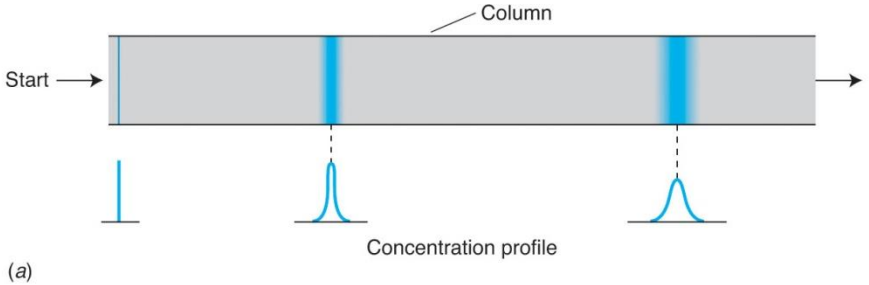
Efficiency of separation

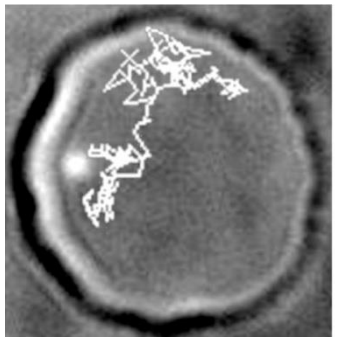


Efficiency of separation

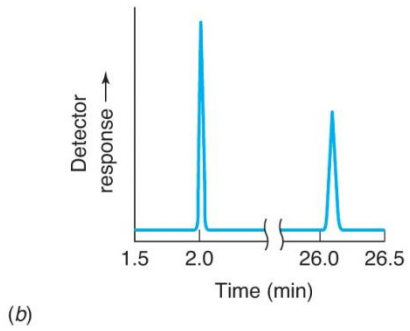
Diffusion: main cause of peak (or band) broadening

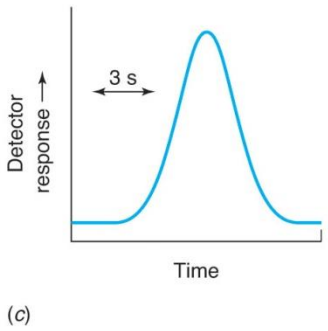
- A band of solute broadens as it moves through a chromatography column
- One main cause of band broadening is *diffusion*, the net transport of a solute from a region of high concentration to a region of low concentration caused by the random movement of molecules

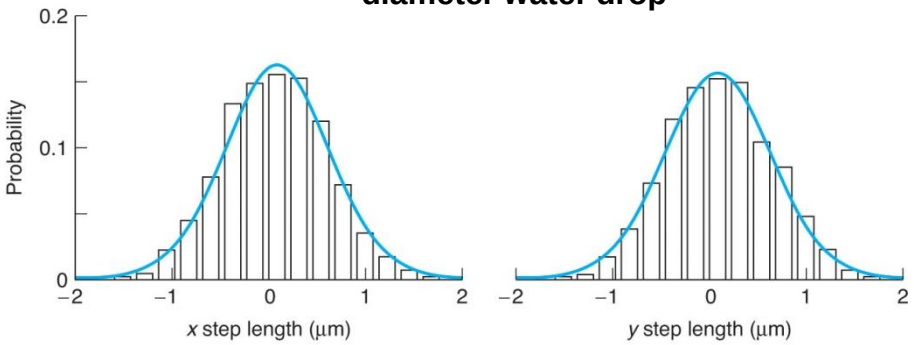




Random motion of molecules: Brownian motion of a 290-nm diameter fluorescent bead in a 20-μm-diameter water drop



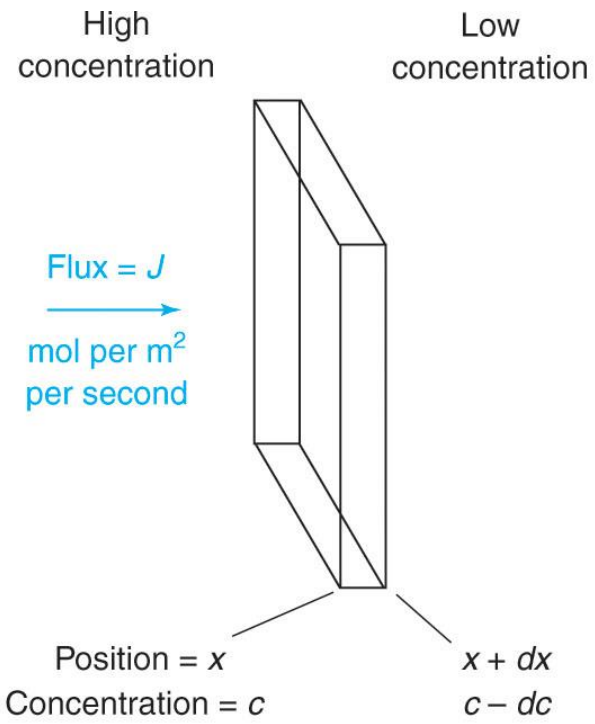




Efficiency of separation

Diffusion: main cause of peak (or band) broadening

Fick's first law of diffusion: the flux (*J*) of molecules diffusing across a plane of unit area is proportional to the concentration gradient and to the diffusion coefficient (*D*)



$$\text{Flux}\left(\frac{\text{mol}}{\text{m}^2 \cdot \text{s}}\right) \equiv J = -D\frac{dc}{dx}$$

- **Flux (*J*): number of moles crossing each square meter per second (unit: $mol.m^{-2}.s^{-1}$)**
- **Diffusion coefficient (D): proportionality constant (unit: $m^2.s^{-1}$)**
- **Negative sign is necessary because the net flux is from the region of high concentration to the region of low concentration**

Efficiency of separation

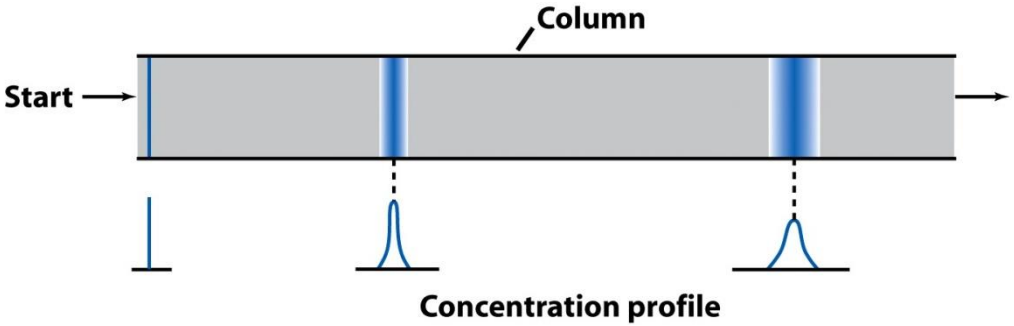
Diffusion: main cause of peak (or band) broadening

TABLE 22-1 Representative diffusion coefficients at 298 K

Solute	Solvent	Diffusion coefficient (m^2/s)
H_2O	H_2O	2.3×10^{-9}
Sucrose	H_2O	0.52×10^{-9}
Glycine	H_2O	1.1×10^{-9}
CH_3OH	H_2O	1.6×10^{-9}
Ribonuclease (FM 13 700)	H_2O (293 K)	0.12×10^{-9}
Serum albumin (FM 66 000)	H_2O (293 K)	0.059×10^{-9}
I_2	Hexane	4.0×10^{-9}
CCl_4	Heptane	3.2×10^{-9}
N_2	CCl_4	3.4×10^{-9}
$CS_2(g)$	Air (293 K)	1.0×10^{-5}
O_2 (g)	Air (273 K)	1.8×10^{-5}
H^+	H_2O	9.3×10^{-9}
OH^-	H_2O	5.3×10^{-9}
Li^+	H_2O	1.0×10^{-9}
Na^+	H_2O	1.3×10^{-9}
K^+	H_2O	2.0×10^{-9}
Cl^-	H_2O	2.0×10^{-9}
I^-	H_2O	2.0×10^{-9}

Efficiency of separation

Diffusion: main cause of peak (or band) broadening



Broadening of chromatography band by diffusion

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

(확률밀도함수)

$$c = \frac{m}{\sqrt{4\pi Dt}} e^{-x^2/(4Dt)}$$

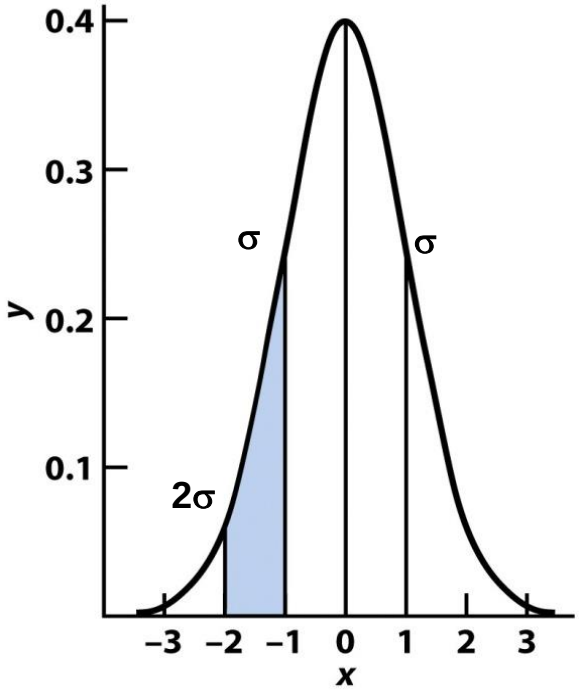
$$c(at\ x = 0) = \frac{m}{\sqrt{4\pi Dt}}$$

$$2\sigma^2 = 4Dt$$

$$\sigma = \sqrt{2Dt}$$

$$\text{Bandwidth (w)} \propto \sigma \propto \sqrt{t}$$

c: solute concentration (mol/m^3)
D: diffusion coefficient
m: amount of solute loaded per unit cross-sectional area of the column in the beginning (mol/m^2)
x: the distance along the column from the current center of the band
t: time post sample loading

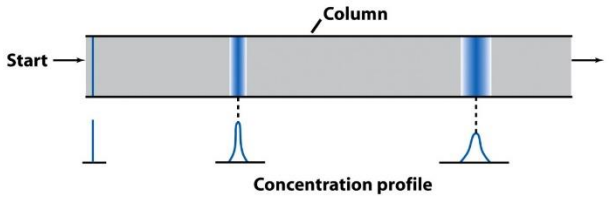


Efficiency of separation

Plate height(이론단의 높이): a measure of column efficiency

$$\sigma = \sqrt{2Dt}$$

$$t = \frac{x}{u_x}$$



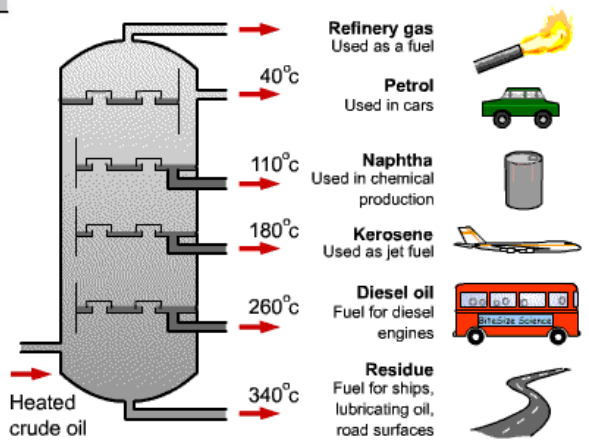
x : distance traveled by a solute,

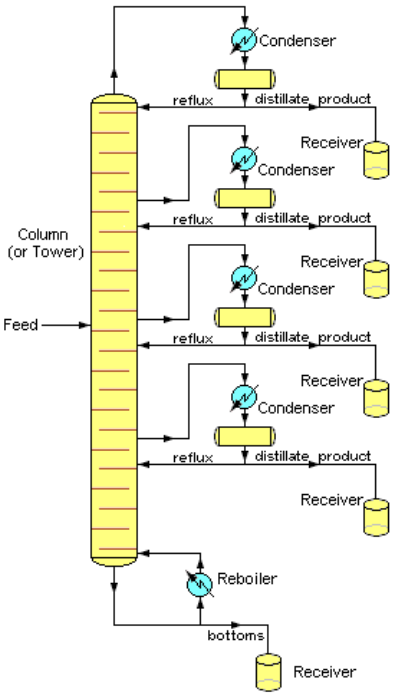
u_x : Linear velocity of the solute

$$\sigma^2 = 2Dt = 2D\frac{x}{u_x} = \left(\frac{2D}{u_x}\right)x = Hx$$

H : plate height

$$H = \frac{2D}{u_x} = \frac{\sigma^2}{x}$$





Continuous fractional distillation

Plate height:

- **Proportionality constant between the variance (σ^2) of the band and the distance (x) the solute has traveled along the column**
- **Also called height equivalent to a theoretical plate (HETP)**
- **Approximately the length of column required for one equilibration of solute between mobile and stationary phases**

Small plate height ⇒ narrow peaks ⇒ high resolution ⇒ better separation

Efficiency of separation

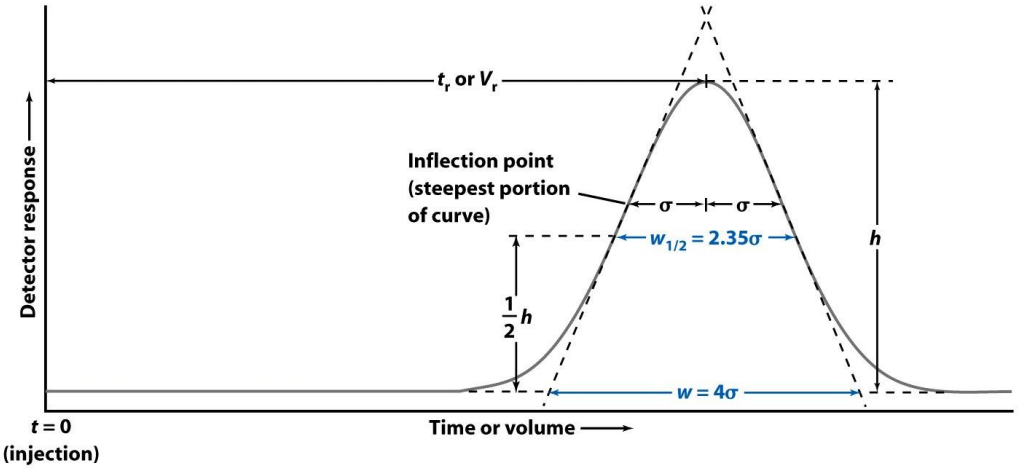
Plate height: a measure of column efficiency

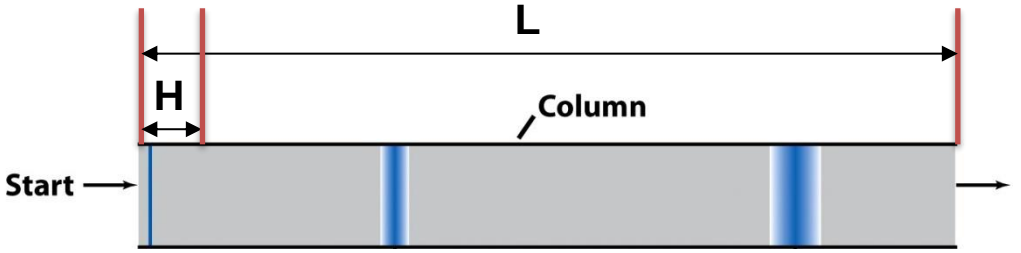
N: number of (theoretical) plates(이론단수) on column

$$N = \frac{L}{H} = \frac{Lx}{\sigma^2} = \frac{L^2}{\sigma^2} = \frac{L^2}{(w/4)^2} = \frac{16L^2}{w^2}$$

$$N = \frac{16{t_r}^2}{w^2} = \frac{{t_r}^2}{\sigma^2}$$

$$N = \frac{16{t_r}^2}{w^2} = \frac{16{t_r}^2}{(1.70w_{1/2})^2} = \frac{5.55{t_r}^2}{{w_{1/2}}^2}$$





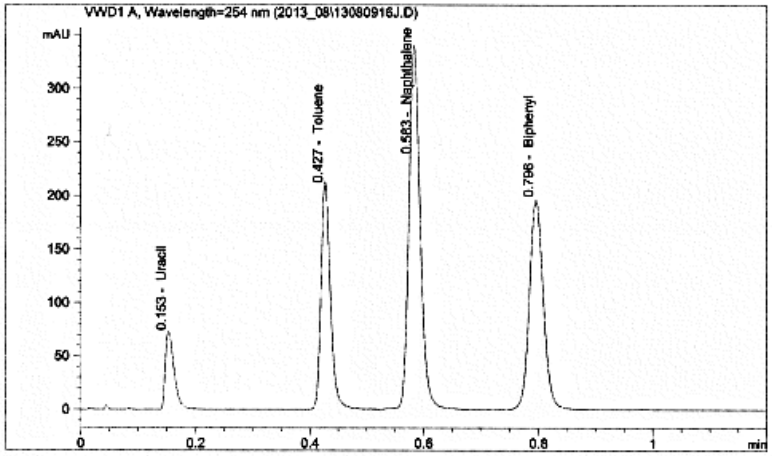
LC Certificate of Analysis

RESTEK

Column Performance Report

Packing Material: Raptor BiPh
Catalog Number : 9309A32
Serial Number : 13080916J
Batch Number : 130403R

Column Length : 30 mm
Inside Diameter: 2.1 mm
Particle Size : 2.7µm



Mobile Phase : 55/45 ACN/Water
Flow Rate : 0.4 mL/min
Sample : Cat# 35074
Pressure : 1090 psi

Wavelength : 254 nm
Sample Amount : 1.0 µl
Temperature : Ambient

Storage Solvent : Methanol

Performance results for last peak:

Retention time	: 0.80 min
k'	: 4.19
Width at 1/2 height	: 0.03 min
Symmetry at 10%	: 1.15
Plates/column	: 5335
Plates/meter	: 177832

Analyst: PKM Tested: 8/27/2013 1290-1

$$H = \frac{L}{N} = \frac{30\ \text{mm}}{5335\ \text{plates}} = 5.6 \times 10^{-3}\ \text{mm/plate}$$

RESTEK QA APPROVED

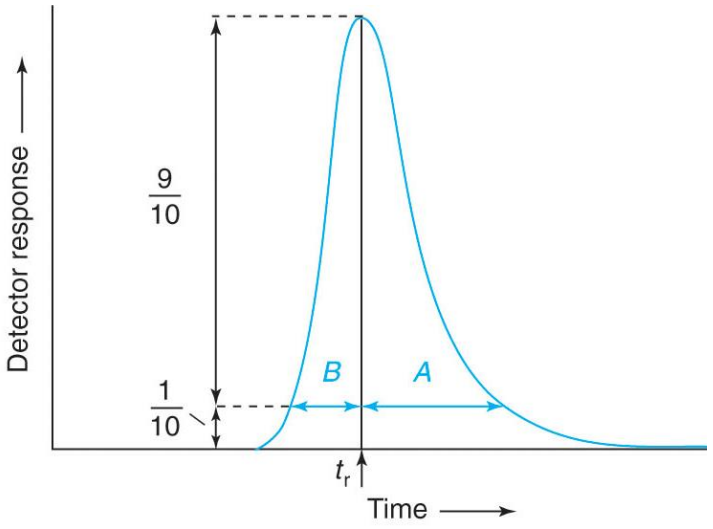
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Efficiency of separation

Plate height: a measure of column efficiency

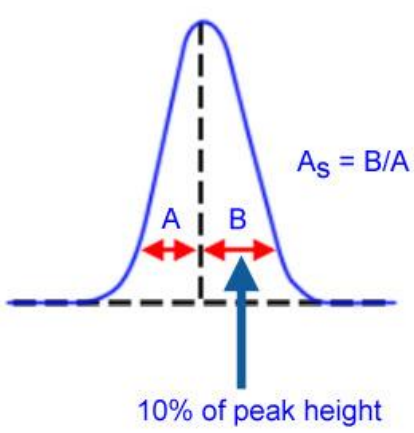
Estimation of the number of plates for an asymmetric tailing peak

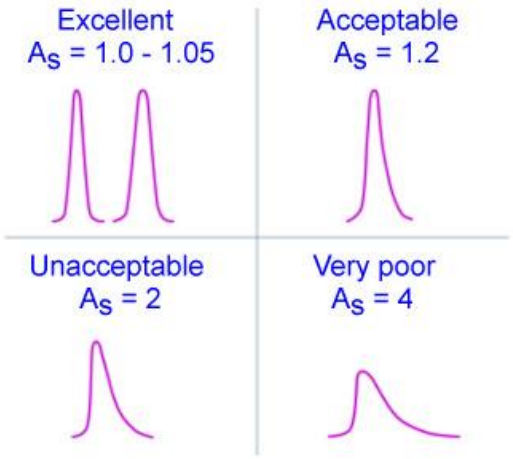


$$N \approx \frac{41.7(t_r/w_{0.1})^2}{A/B + 1.25}$$

$w_{0.1}$ = A+B

A/B: Asymmetry factor





A_s: Asymmetry factor

Efficiency of separation

Factors affecting resolution

$$\text{Resolution} = \frac{\sqrt{N}}{4}(\gamma - 1)$$

$\gamma = \frac{u_A}{u_B} = \frac{t_B}{t_A}$: unadjusted relative retention ($\gamma > 1$)

u_A, u_B: linear velocity of components A and B

t_A, t_B: retention times of components A and B

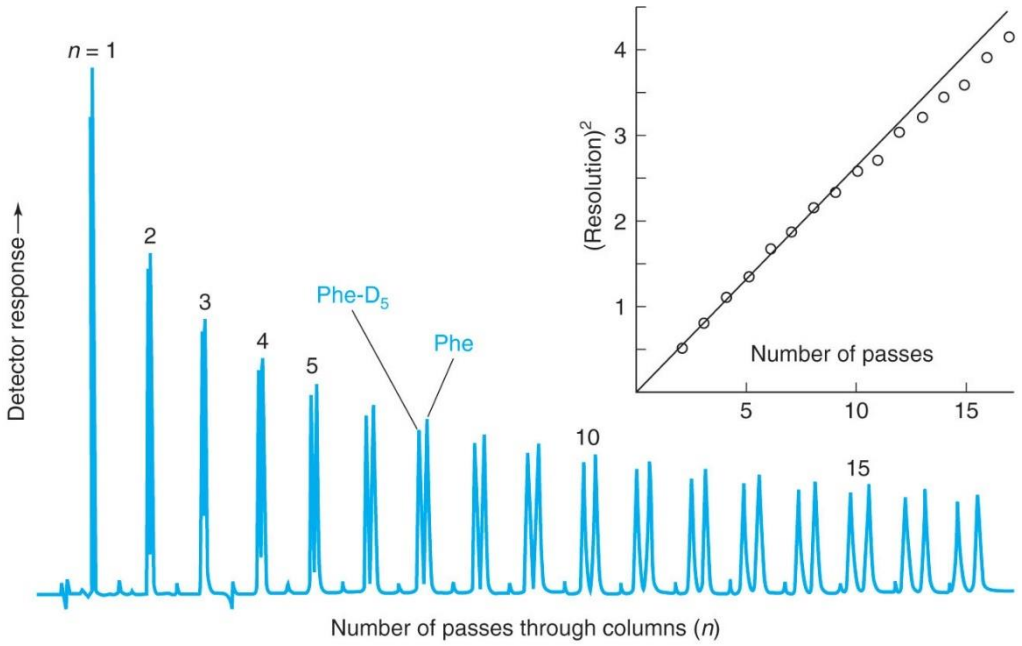
$$\text{Resolution} \propto \sqrt{N} \propto \sqrt{L} \propto \frac{1}{\sqrt{H}}$$

N: number of plate of peak A

Q: Peak width (w)와 H 및 N과의 상관관계는?

$$w \propto \sigma \propto \sqrt{H} \propto \frac{1}{\sqrt{N}}$$

Separation of 0.5 mM L-Phe and 0.5 mM L-Phe-D_5 by repeated passes through a pair of Hypersil C8 columns (25 cm x 4.6 mm) eluted with 10:90 acetonitrile:water containing 25 mM Na_2SO_4 and 0.1% TFA in the water. The relative retention on the first pass is α=1.03.



Why bands spread

Plate height equation: The van Deemter equation

- **Plate height, *H*, is proportional to the variance of a chromatographic band**
- **The van Deemter equation tells us how the column and flow rate affect plate height**

van Deemter equation for plate height :

$$H \approx A + \frac{B}{u_x} + C \cdot u_x$$

A : Multiple flow paths

$\frac{B}{u_x}$: Longitudinal diffusion

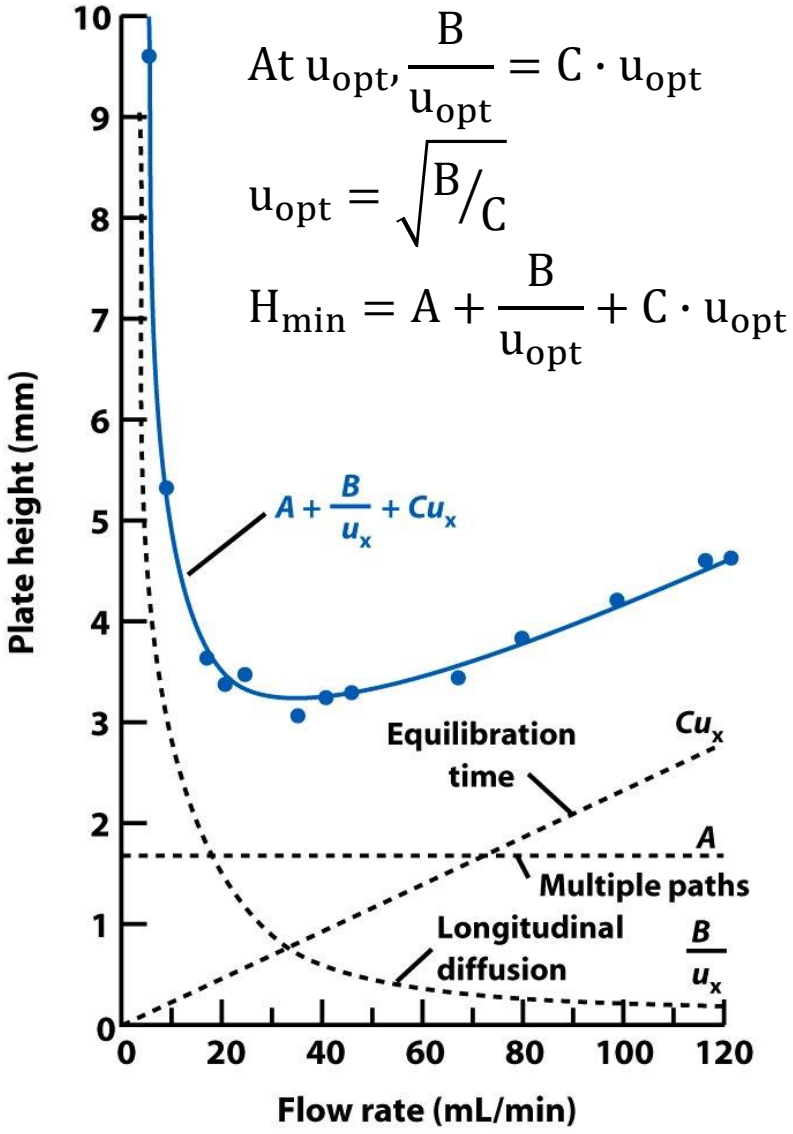
$C \cdot u_x$: Equilibration time

A, B, C : constants for a given column and stationary phase

$$\text{At } u_{opt}, \frac{B}{u_{opt}} = C \cdot u_{opt}$$

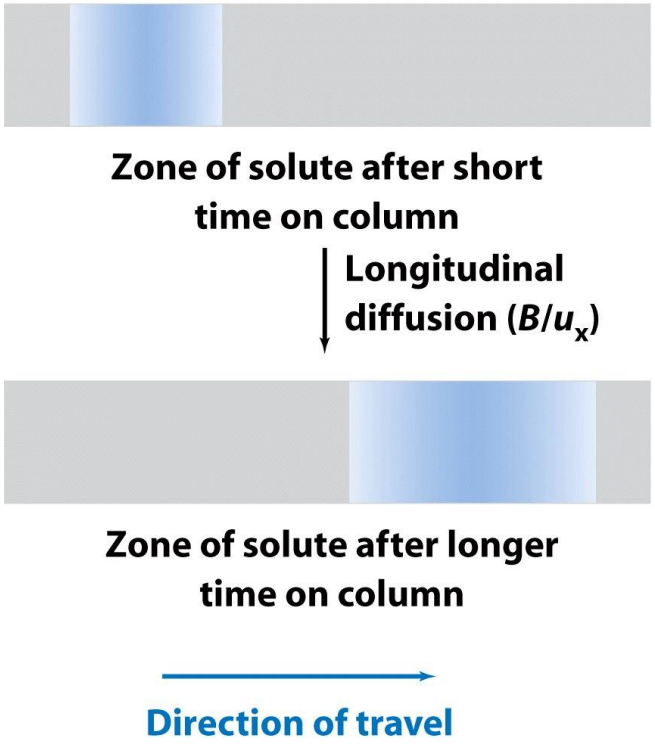
$$u_{opt} = \sqrt{B/C}$$

$$H_{min} = A + \frac{B}{u_{opt}} + C \cdot u_{opt}$$



Why bands spread

Longitudinal diffusion



van Deemter equation for plate height:

$$H \approx A + \frac{B}{u_x} + C \cdot u_x$$

A: Multiple paths

$\frac{B}{u_x}$: Longitudinal diffusion

$C \cdot u_x$: Equilibration time

The faster the linear flow
$\Rightarrow$ The less time spent in the column
$\Rightarrow$ The less diffusion
$\Rightarrow$ The less band broadening

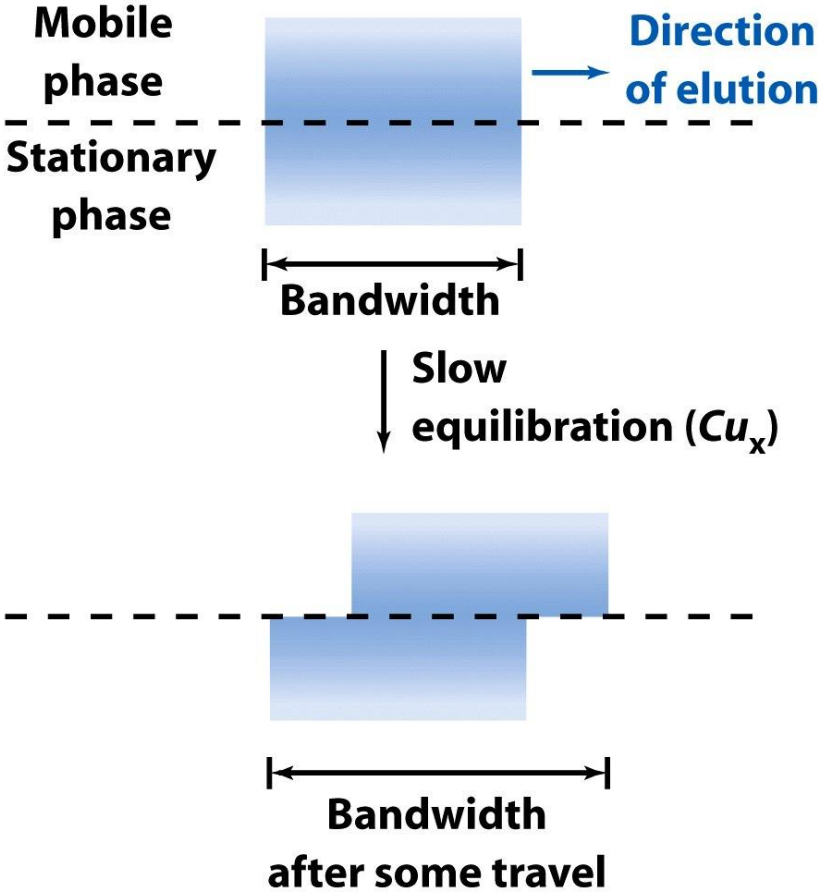
$$\sigma^2 = 2D_m t = \frac{2D_m L}{u_x}$$

$$H_D = \frac{\sigma^2}{L} = \frac{2D_m}{u_x} \equiv \frac{B}{u_x}$$

D_m: diffusion coefficient in the mobile phase
H_D: plate height due to longitudinal diffusion
L: column length
u_x: linear flow rate
t: time

Why bands spread

Finite equilibration time between phases



van Deemter equation for plate height:

$$H \approx A + \frac{B}{u_x} + C \cdot u_x$$

A: Multiple paths

$\frac{B}{u_x}$: Longitudinal diffusion

$C \cdot u_x$: Equilibration time

The slower the linear flow
$\Rightarrow$ The more complete equilibration is
$\Rightarrow$ The less band broadening

Why bands spread

Finite equilibration time between phases

Plate height due to finite equilibration time (mass transfer term):

$$H_{masstransfer} = Cu_x = (C_s + C_m)u_x$$

C_s: mass transfer through the stationary phase
C_m: mass transfer through the mobile phase

$$C_s = \frac{2k}{3(k+1)^2}\frac{d^2}{D_s}$$

$$C_m = \frac{1+6k+11k^2}{24(k+1)^2}\frac{r^2}{D_m}$$

k: retention factor
d: thickness of the stationary phase
D_s: diffusion coefficient in the stationary phase
r: column radius
D_m: diffusion coefficient in the mobile phase

- **Decreasing *d* reduces plate height and increases efficiency because solute can diffuse faster from the stationary phase into the mobile phase**
- **Decreasing *r* reduces plate height by decreasing the distance through which solute must diffuse to reach the stationary phase**
- **Increasing temperature reduces plate height by increasing D_s**

Why bands spread

Finite equilibration time between phases

- Increasing the temperature allows the linear flow rate to be increased while maintaining acceptable resolution
- Resolution is maintained because of the increased rate of mass transfer between phases at elevated temperature

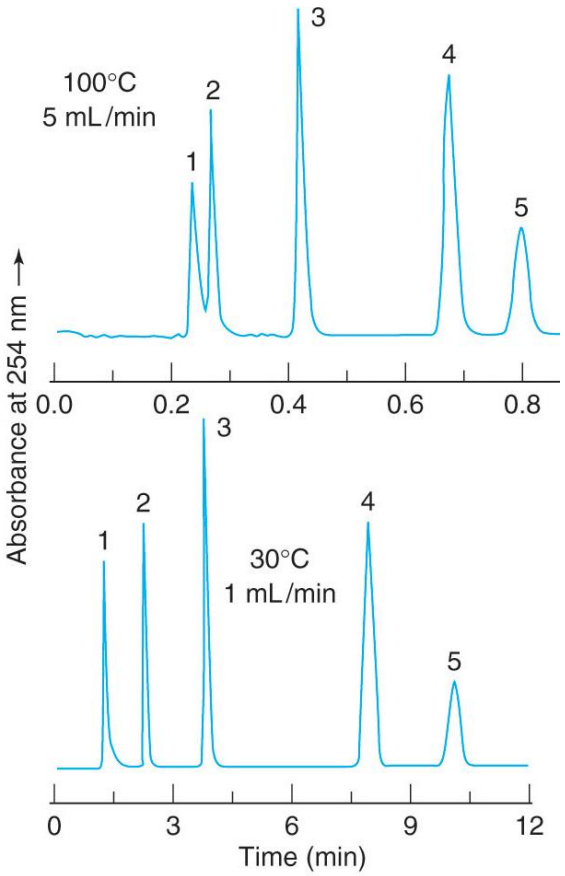
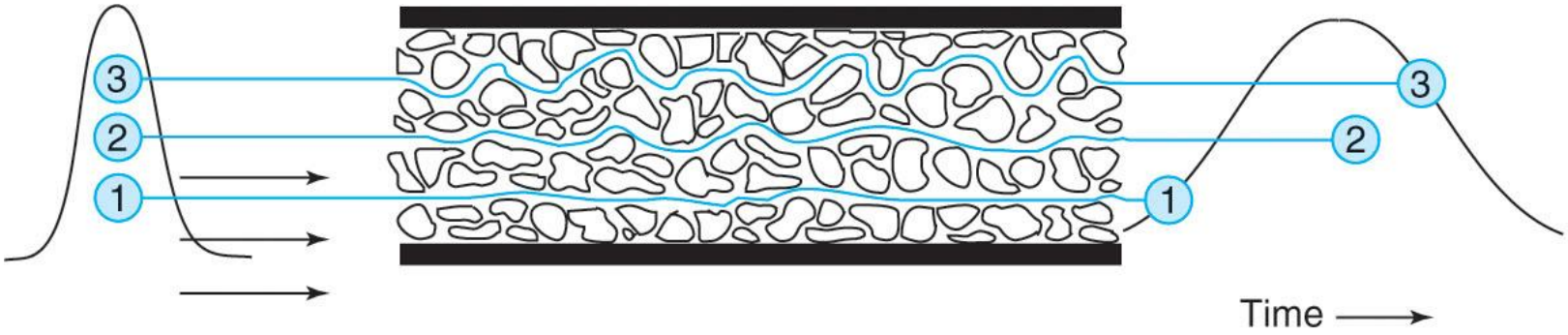


FIGURE 22-19 Liquid chromatography showing decreased analysis time when temperature is raised from 30° to 100°C. 1, uracil; 2, *p*-nitroaniline; 3, methyl benzoate; 4, phenetole; 5, toluene. The 4.6-mm-diameter × 100-cm-long column was packed with 4.5-μm-diameter zirconia (ZrO_2) coated with 2.1 wt% polybutadiene and eluted with 20 vol% acetonitrile in water. [From J. Li, Y. Hu, and P. W. Carr, "Fast Separations at Elevated Temperatures on Polybutadiene-Coated Zirconia Reversed-Phase Material," *Anal. Chem.* **1997,** *69*, 3884.] For a silica-based stationary phase, temperature is usually kept below 60°C to prevent hydrolysis of the silica.

Why bands spread

Multiple flow paths



The smaller the stationary particles, the less serious this problem is

Q: What is the value of A in van Deemter equation for an open tubular column?

van Deemter equation for plate height :

$$H \approx A + \frac{B}{u_x} + C \cdot u_x$$

A : Multiple paths

$\frac{B}{u_x}$: Longitudinal diffusion

$C \cdot u_x$: Equilibration time

Why bands spread

Advantages of open tubular columns vs. packed columns

- Lower pressure
- Higher linear flow rate $\Rightarrow$ lower longitudinal diffusion $\Rightarrow$ smaller plate height $\Rightarrow$ narrower peak $\Rightarrow$ higher resolution
- No multiple path flow $\Rightarrow$ smaller plate height $\Rightarrow$ narrower peak $\Rightarrow$ higher resolution
- Longer column length $\Rightarrow$ higher number of plate $\Rightarrow$ higher resolution
- Higher linear flow rate $\Rightarrow$ shorter retention time $\Rightarrow$ shorter analysis time

TABLE 22-3 Comparison of packed and wall-coated open tubular column performance[a]

Property	Packed	Open tubular
Column length, L	2.4 m	100 m
Linear gas velocity	8 cm/s	16 cm/s
Plate height for methyl oleate	0.73 mm	0.34 mm
Retention factor, k, for methyl oleate	58.6	2.7
Theoretical plates, N	3 290	294 000
Resolution of methyl stearate and methyl oleate	1.5	10.6
Retention time of methyl oleate	29.8 min	38.5 min

a. Methyl stearate ($CH_3(CH_2)_{16}CO_2CH_3$) and methyl oleate (cis-$CH_3(CH_2)_7CH{=}CH(CH_2)_7CO_2CH_3$) were separated on columns with poly(diethylene glycol succinate) stationary phase at 180°C.